

Throughput

Throughput means how much data can be transferred from one location to another in a given amount of time. It is used to measure the performance of communication networks, hard drives and RAM. Throughput can also be termed as the average successful message delivery on a communications network. Throughput is usually measured in bits per second, bytes per second, or number of data packets per second.

Goodput

Goodput is relevant in telecommunication products, where we implement many layers of protocol, and at each protocol layer there is a header and footer added to the data. Goodput is the application-level throughput—the number of useful bits, per unit of time, forwarded by the network from a source node to the destination node, excluding protocol overhead and retransmitted data packets. Goodput refers to the measurement of actual data successfully transmitted from the senders to the receivers; the throughput of data that we get at the application layer could be 60 per cent of the total throughput.

Latency

Latency is a measure of the time delay experienced in a system or product due to many factors. Latency or delay (see Figures 8 and 9) is categorised into hardware latency, software latency, and mechanical latency.

Hardware latency is the time delay introduced in the printed circuit board (PCB). This consists of the hardware propagation delay of the semiconductor used in the chip, the propagation delay in the buses or the copper wire used in the PCBs, and the delay in the firmware burnt on the chip. All these three delays are responsible for introducing hardware latency. A product architect must certainly keep an eye on hardware latency.

Software latency is the latency introduced by software, and is divided into *node latency* and *network latency*. Node latency consists of operating system latency, software stack propagation delay, and the application software execution delay. Operating system latency is further split into *interrupt latency*, *task pre-emption time* and *context switch time*. We will discuss all the operating system latencies separately, and in more detail in a subsequent article.

In addition to the latencies that are particular to a node, we also have network latency coming into the picture if we have a network of nodes, such as a telecommunication network. Some of the causes of network latency are given in Figure 9.

Mechanical latency: Any mechanical work encounters restrictions and delays, which can be modeled by Newtonian physics. For example, the working of disk drives, tape drives and DVDs involves mechanical latency. Here, it is the time needed for the data encoded on a platter to rotate from its

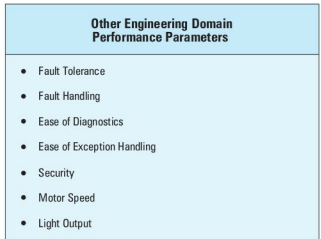


Figure 6: Other engineering domain performance parameters

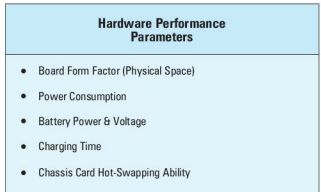


Figure 7: Hardware performance parameters

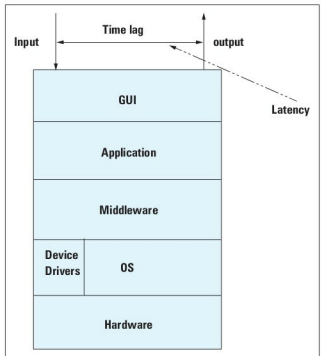


Figure 8: Latency